Information Visualization

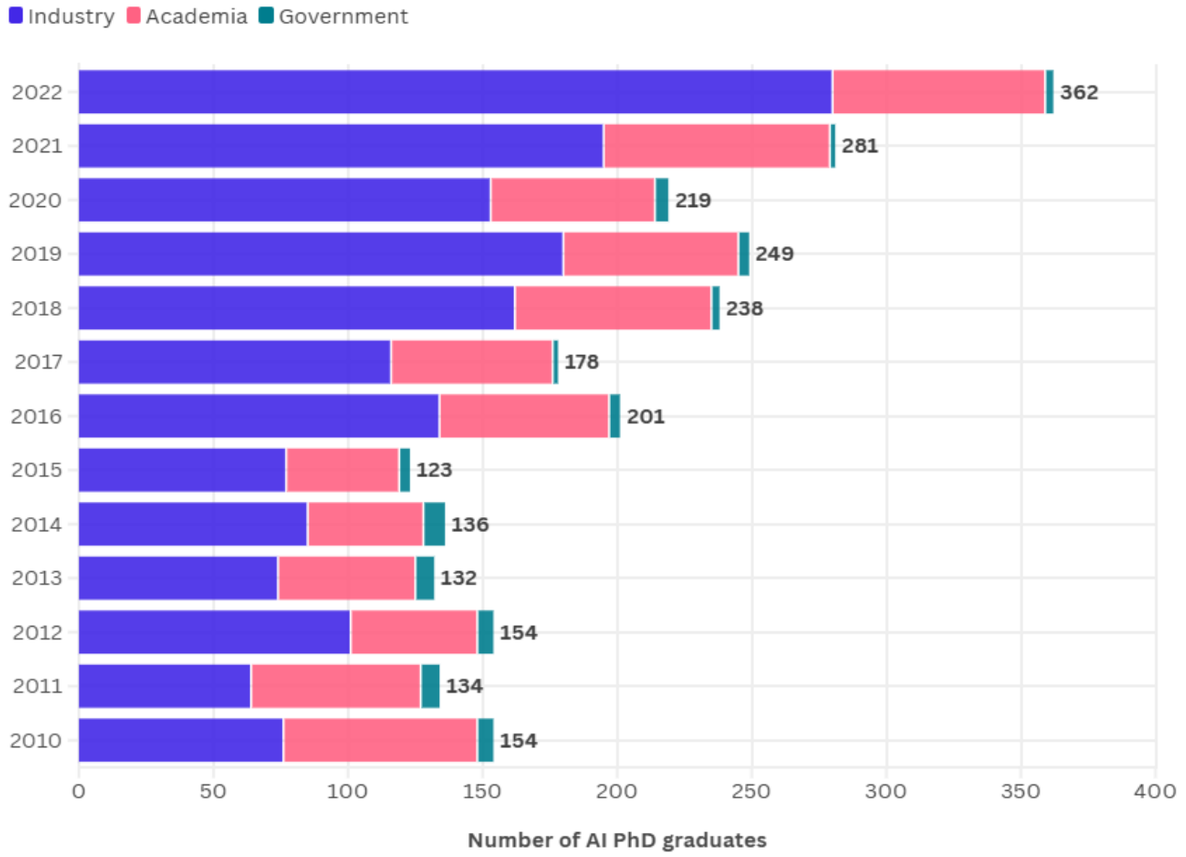
Theme 3. Interaction Techniques

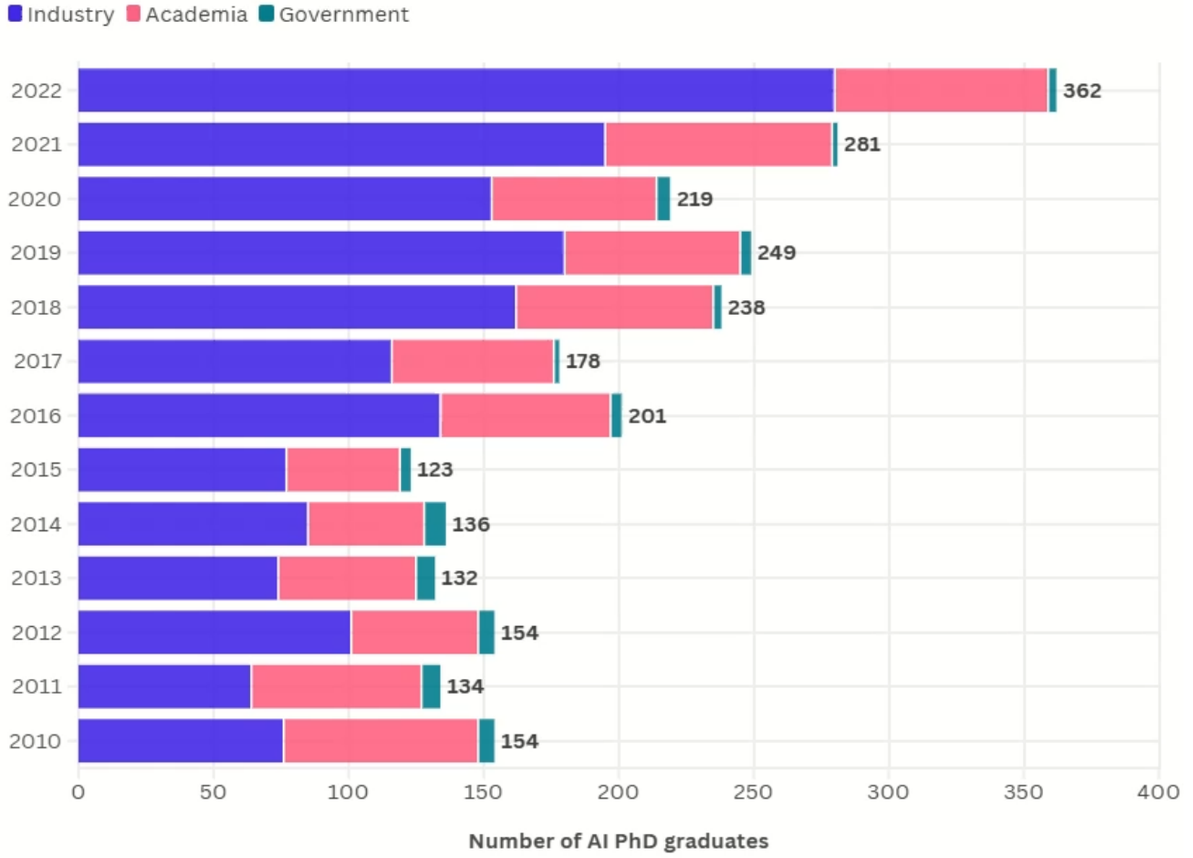
Learning objectives

- Understand why interaction matters for visualization effectiveness.
- Know main interaction design principles (rules of thumb).
- Master common interaction techniques (zoom, pan, filter, brush, details-on-demand, linking).
- Learn how to coordinate multiple views and guidelines.

Why interaction?

- Static pictures are limited; interaction turns a visualization into an exploration tool.





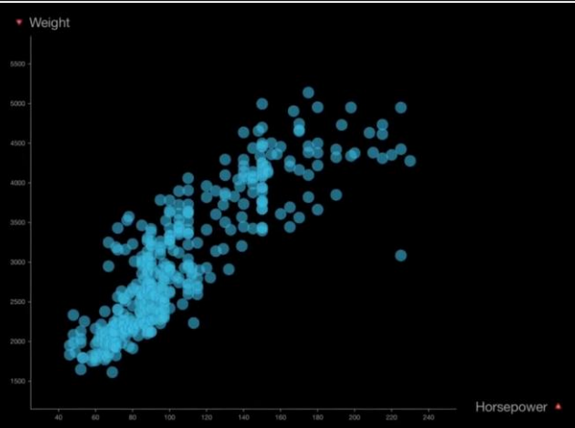
https://flourish.studio/visualisations/bar-charts/

Core interaction techniques

- Selection & highlighting (click, lasso)
- Brushing & linking (brush a subset → highlight across views)
- Zooming & panning (focus & context, semantic zoom)
- Filtering & querying (range sliders, facet filters)
- Details-on-demand (tooltips, side panel)
- Reorder/sort and aggregation controls

Core interaction techniques

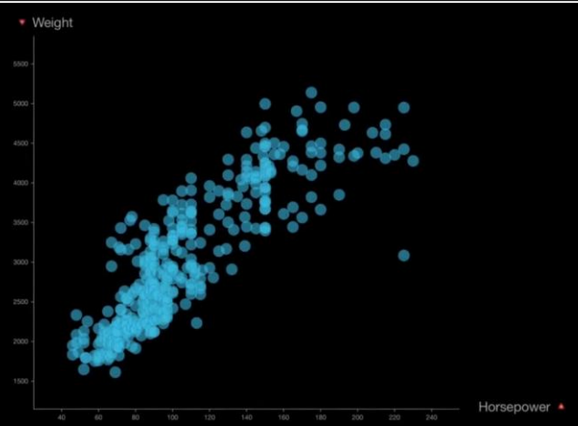
- Lasso Selection



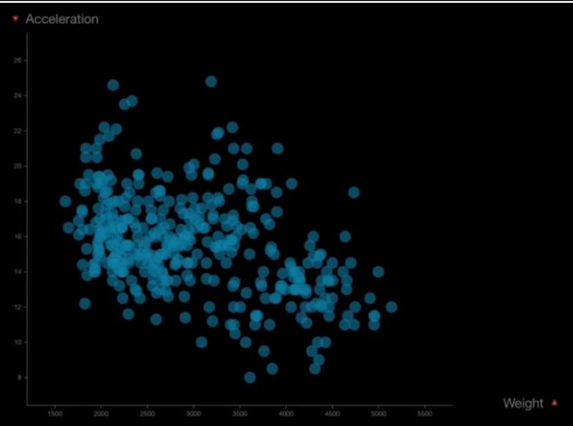
https://ramiksadana.com/video/tangere/select.webm

Core interaction techniques

- Lasso Selection



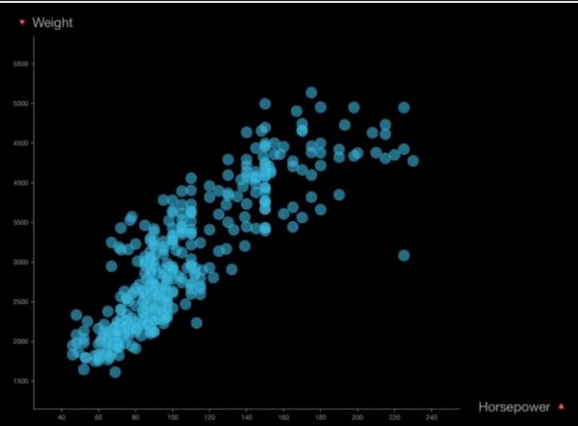
- Marquee Selection



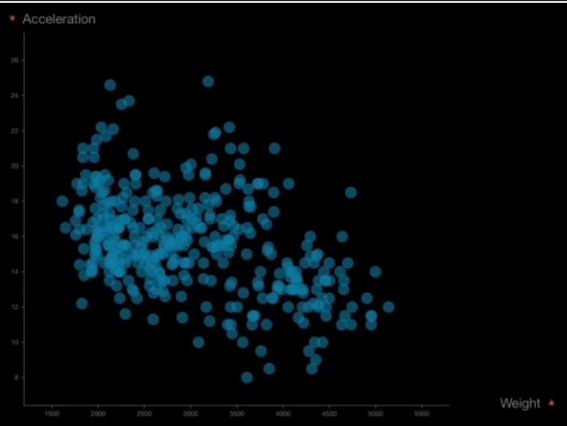
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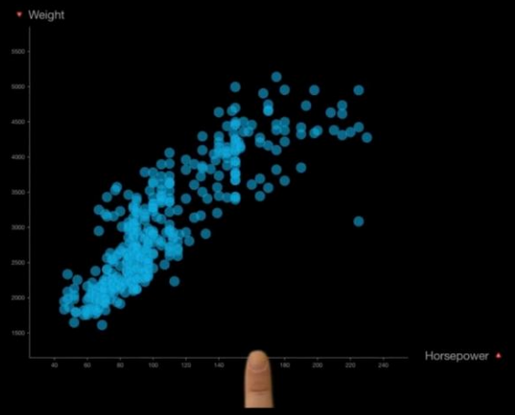
Core interaction techniques

- Lasso Selection



- Marquee Selection



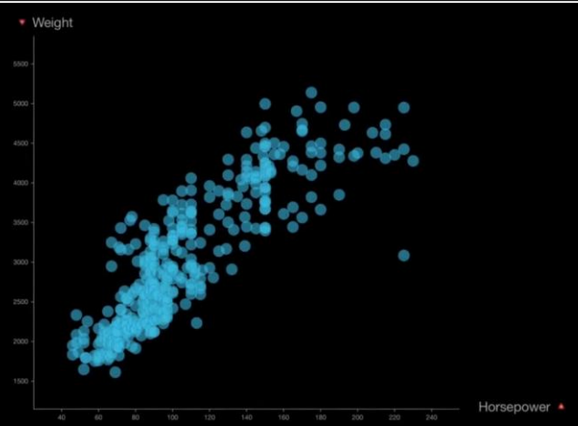


- Axis-drag based Selection

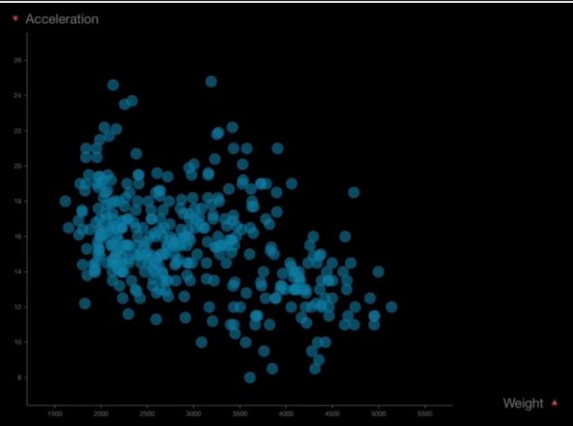
https://ramiksadana.com/video/tangere/select.webm

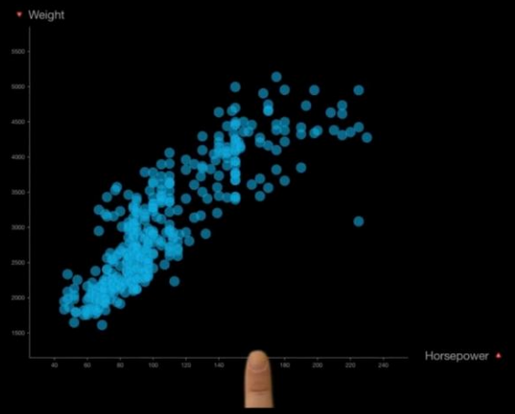
Core interaction techniques

- Lasso Selection

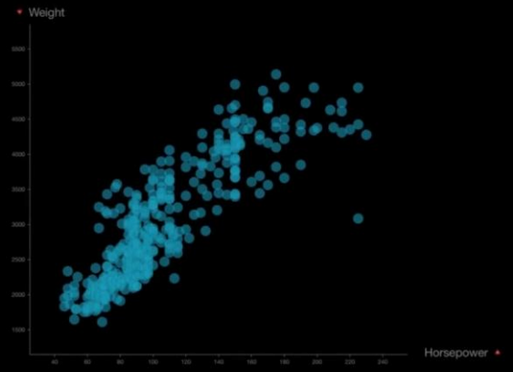


- Marquee Selection





- Axis-drag based Selection



- Off-centered Pointer

https://ramiksadana.com/video/tangere/select.webm

Munzner's Rules of Thumb

- No unjustified 3D
- No unjustified 2D
- Eyes beat memory
- Resolution over immersion
- Overview first, zoom and filter, details on demand
- Responsiveness is required
- Get It Right in Black an White
- Function first, form next

Munzner's Rules of Thumb

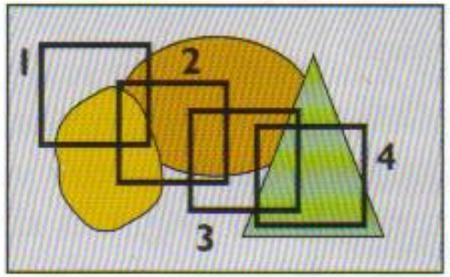
- Overview first, zoom and filter, details on demand



https://maps.google.com/

Zoom & Pan

- Panning
  - is the smooth movement of a viewing frame over a 2D image



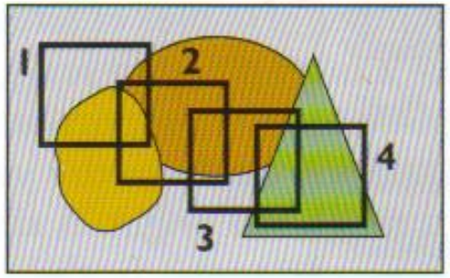
Panning is the smooth movement of a viewing frame over a 2D image



[Spe:130ff]

Zoom & Pan

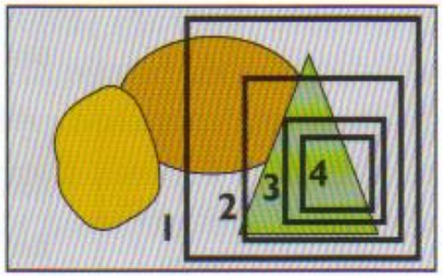
- Panning
  - is the smooth movement of a viewing frame over a 2D image



Panning is the smooth movement of a viewing frame over a 2D image



- Zooming
  - is the increasing magnification of a decreasing fraction of an image (or vice versa)



Zooming is the increasing magnification of a decreasing fraction of an image (or vice versa)

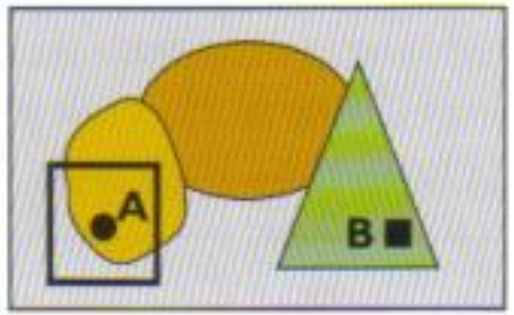
[Spe:130ff]



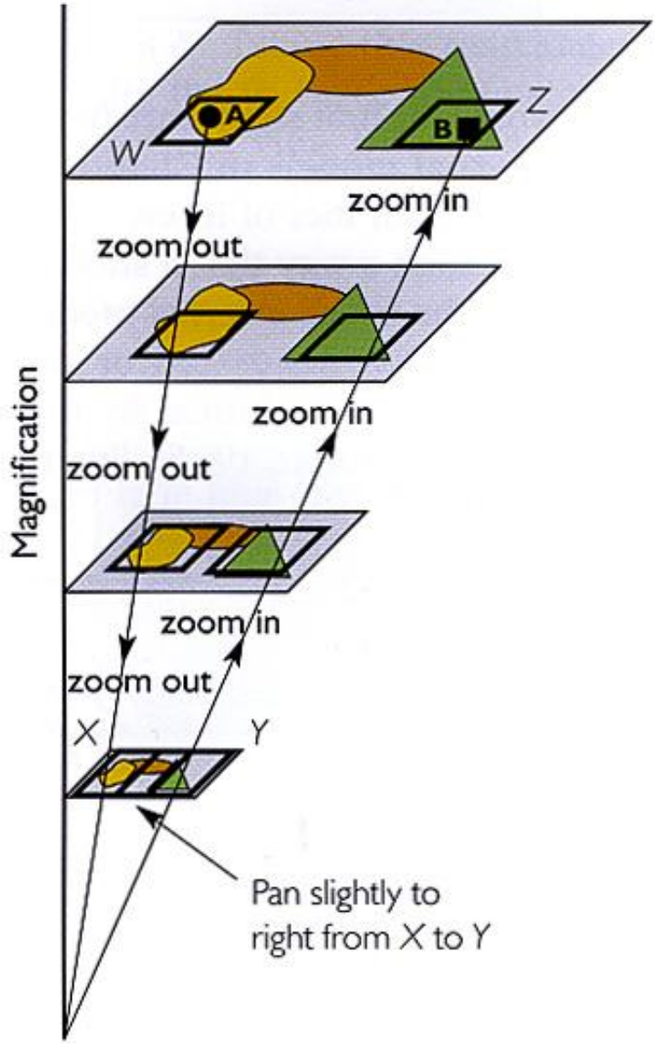
https://maps.google.com/

Zoom & Pan

- Scenario: A user analyzes the focus point A, then he/she wishes to analyze point B.



- To preserve the good overview, he/she zooms out in order to see both points. Then, he/she zooms in to point B
- → Space-*Scale-Diagram*



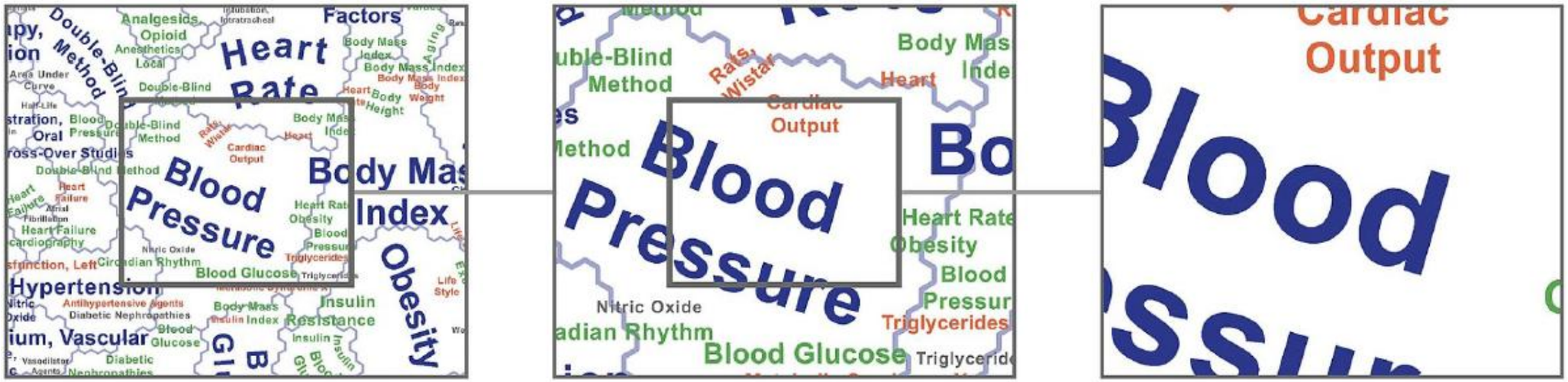
[Furnas, Bederson, Space-Scale Diagrams: Understanding Multi-Scale Interfaces, CHI'95 Proceedings, ACM, 1995, 234-241]

Zoom & Pan

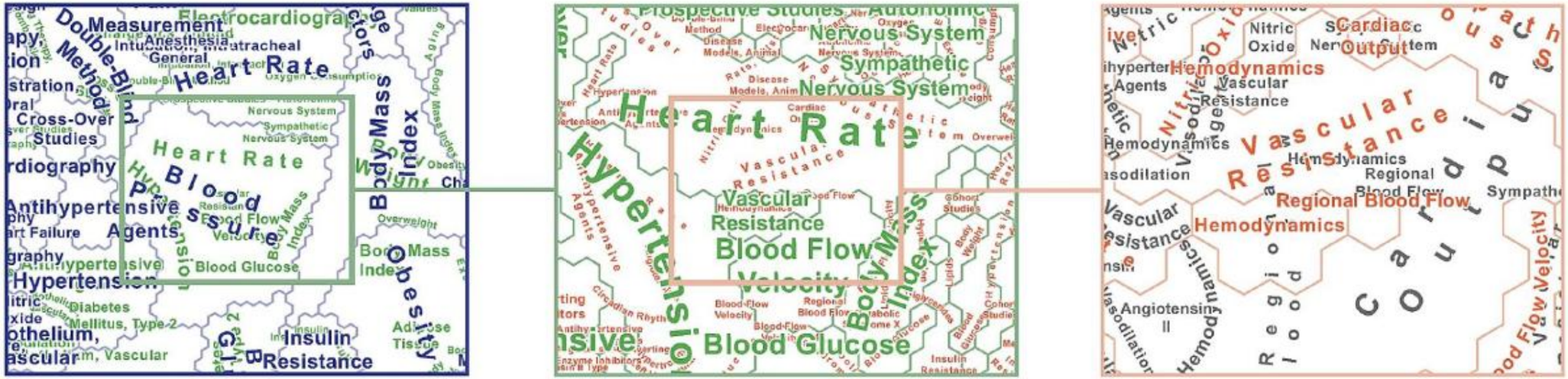

Italy
Black Sea
Bulgaria
Georgia
Portugal
Barcelona
İstanbul
Spain
Tyrrhenian Sea
Greece
Türkiye
Athens
Αθήνα
Tunis
تونس
Syria
Mediterranean Sea
Tunisia
Lebanon
Iraq
Morocco
Israel
Jordan
Algeria
Libya
Egypt
Western
Sahara
Saudi Ara
Jeddah
جدة
Mauritania
Mali
Niger
Sudan
Eritrea
Senegal
Chad

- Geometric Zoom:

https://www.researchgate.net/publication/236105790_Visualizing_the_Topical_Structure_of_the_Medical_Sciences_A_Self-Organizing_Map_Approach

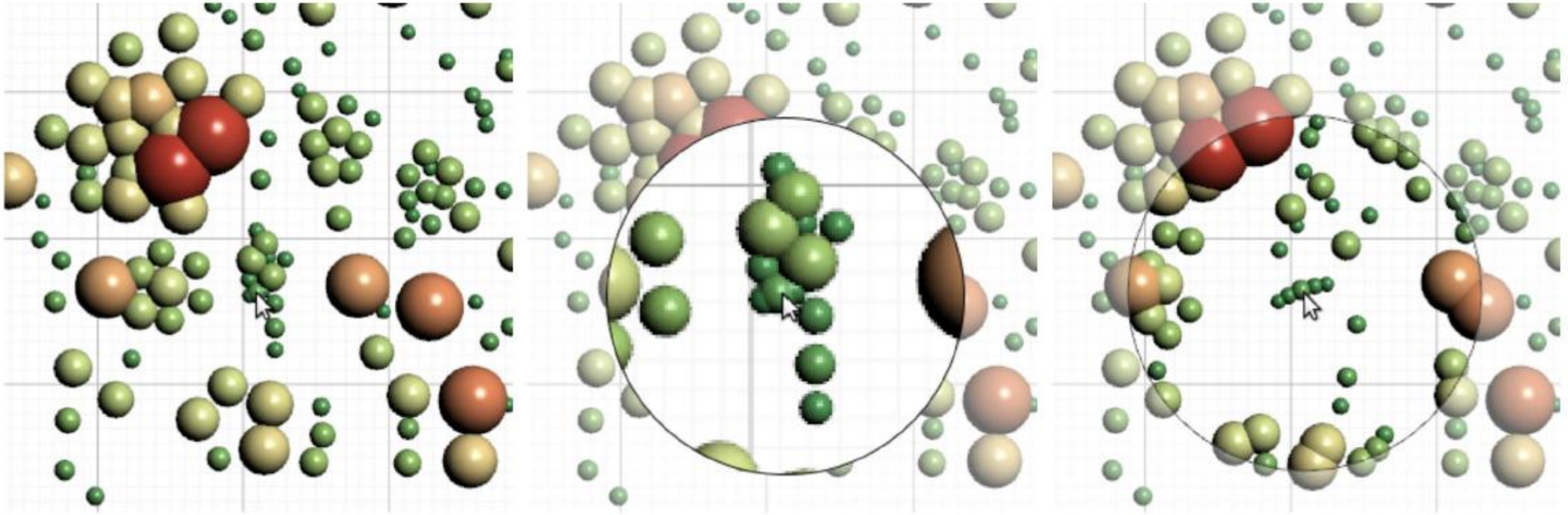


- Semantic Zoom:



[Inspired by Bach's lecturer]

- Zoom using different Lens:



(a) Regular visualization (b) Simple magnification (c) Fish-eye distortion

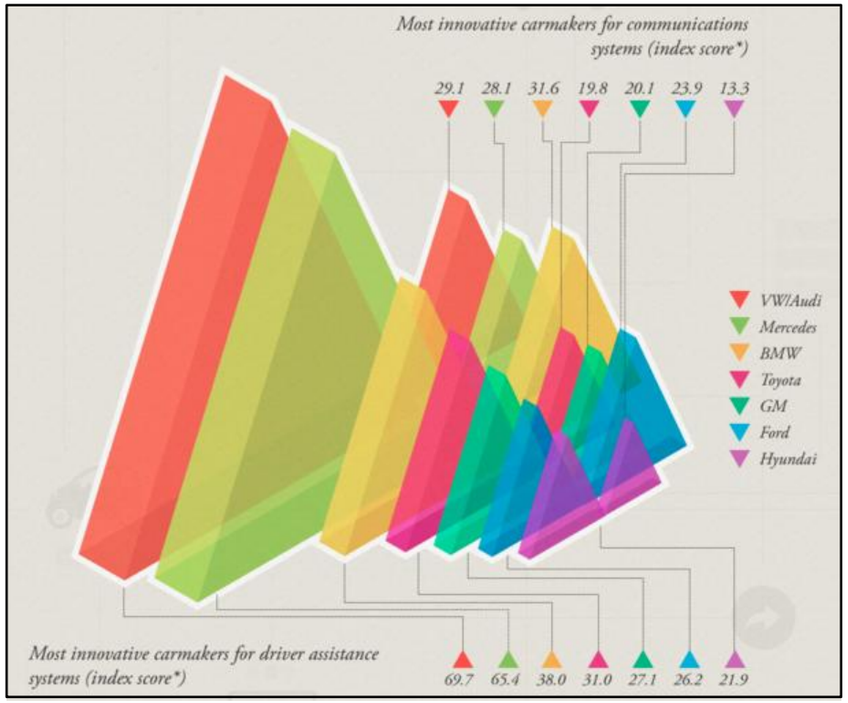
Tominski, C., Gladisch, S., Kister, U., Dachselt, R. and Schumann, H. (2017), Interactive Lenses for Visualization: An Extended Survey. Computer Graphics Forum, 36: 173-200. https://doi.org/10.1111/cgf.12871

[Inspired by Bach's lecturer]

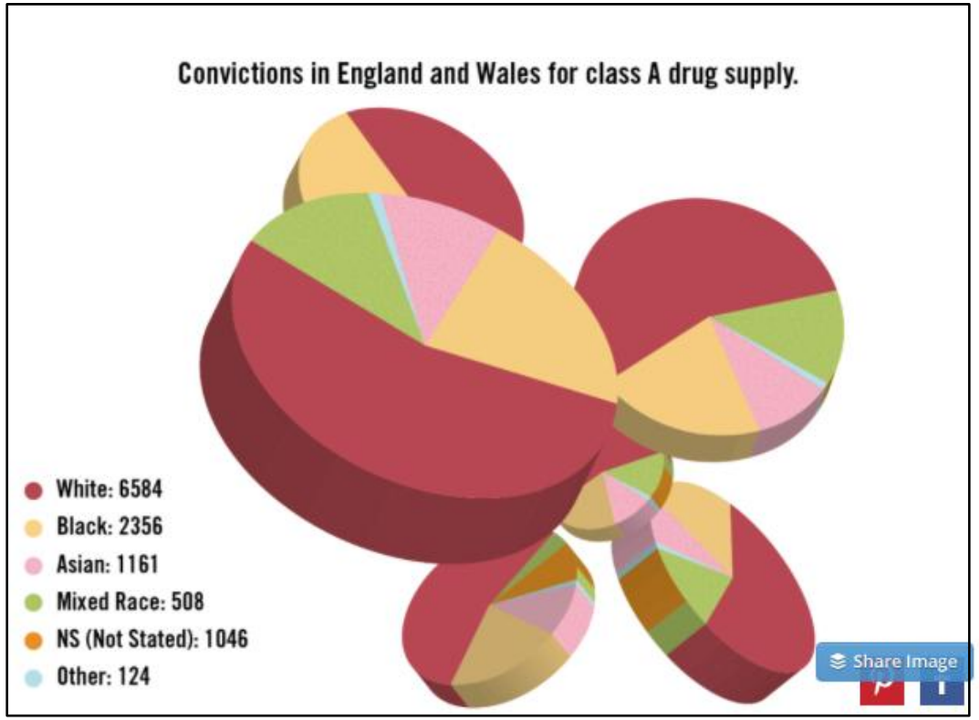
Munzner's Rules of Thumb

- No unjustified 3D
  - Avoid 3D unless the data is inherently three-dimensional.
  - Extra dimension often distorts perception (occlusion, perspective bias).

https://viz.wtf/post/137826497077/eye-popping-3d-triangles



https://viz.wtf/post/139002022202/designer-drugs-ht-ducqn



Munzner's Rules of Thumb

- Responsiveness is required
  - Interactive visualizations must respond quickly enough to keep users "in flow."
  - Delays break attention and exploration.
  - How fast must interaction be?
    - Interaction is a *real time process*, i.e., on the one hand, it must be fast enough to avoid frustration of the user. On the other hand, it should not be too fast to avoid overlooking important things
  - There are three levels:
    - 0.1 second – animations, sliders, sound, …
    - 1 second – system status messages, dialogs, …
    - 10 seconds – cognitive responses (editor, pre-processing, …)

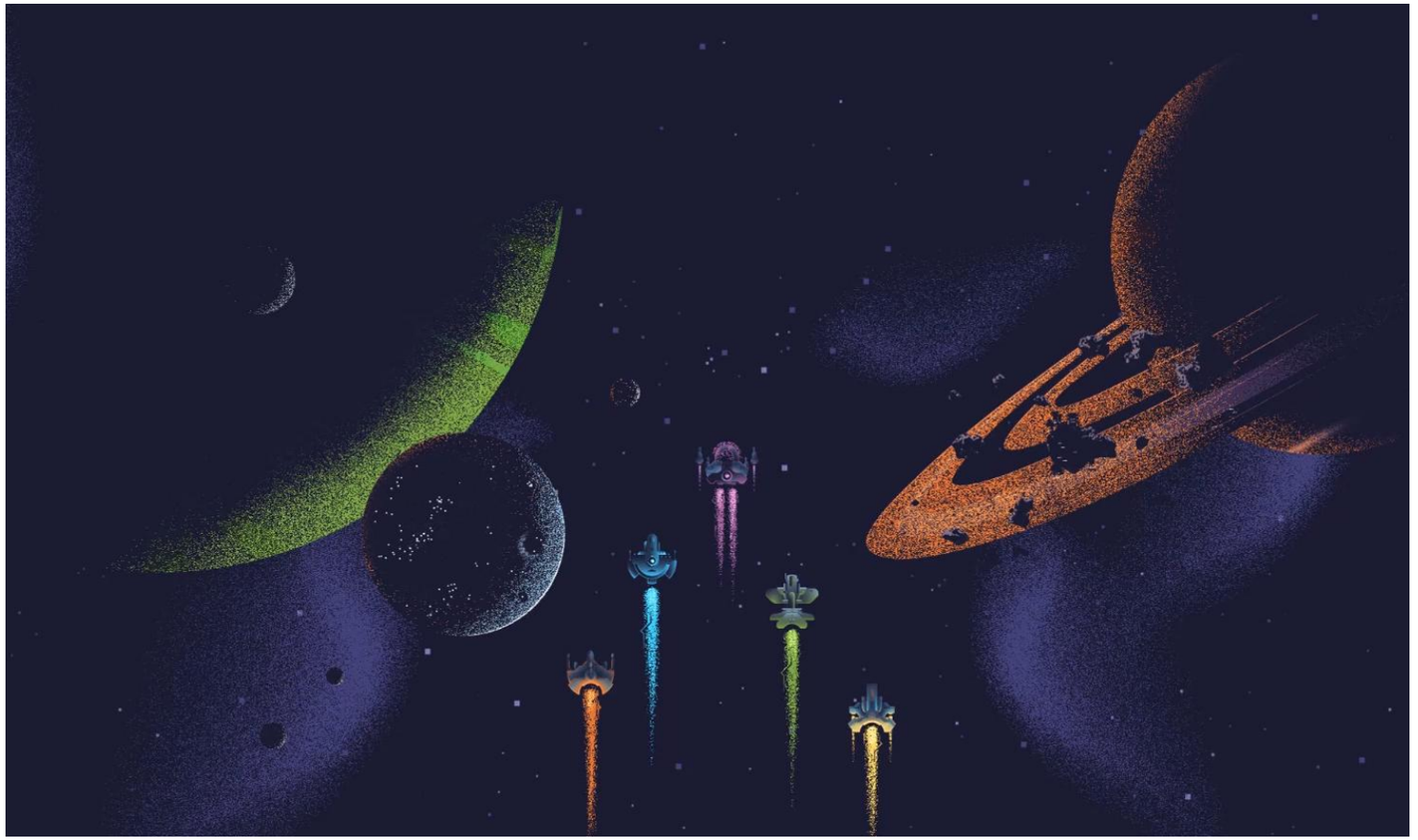
Munzner's Rules of Thumb

- Responsiveness is required
  - If your system has unavoidable delays (e.g., large datasets), what strategies could you use to preserve responsiveness?

Munzner's Rules of Thumb

- Responsiveness is required
  - If your system has unavoidable delays (e.g., large datasets), what strategies could you use to preserve responsiveness?
    - Progressive Loading
    - Previews
    - Sampling



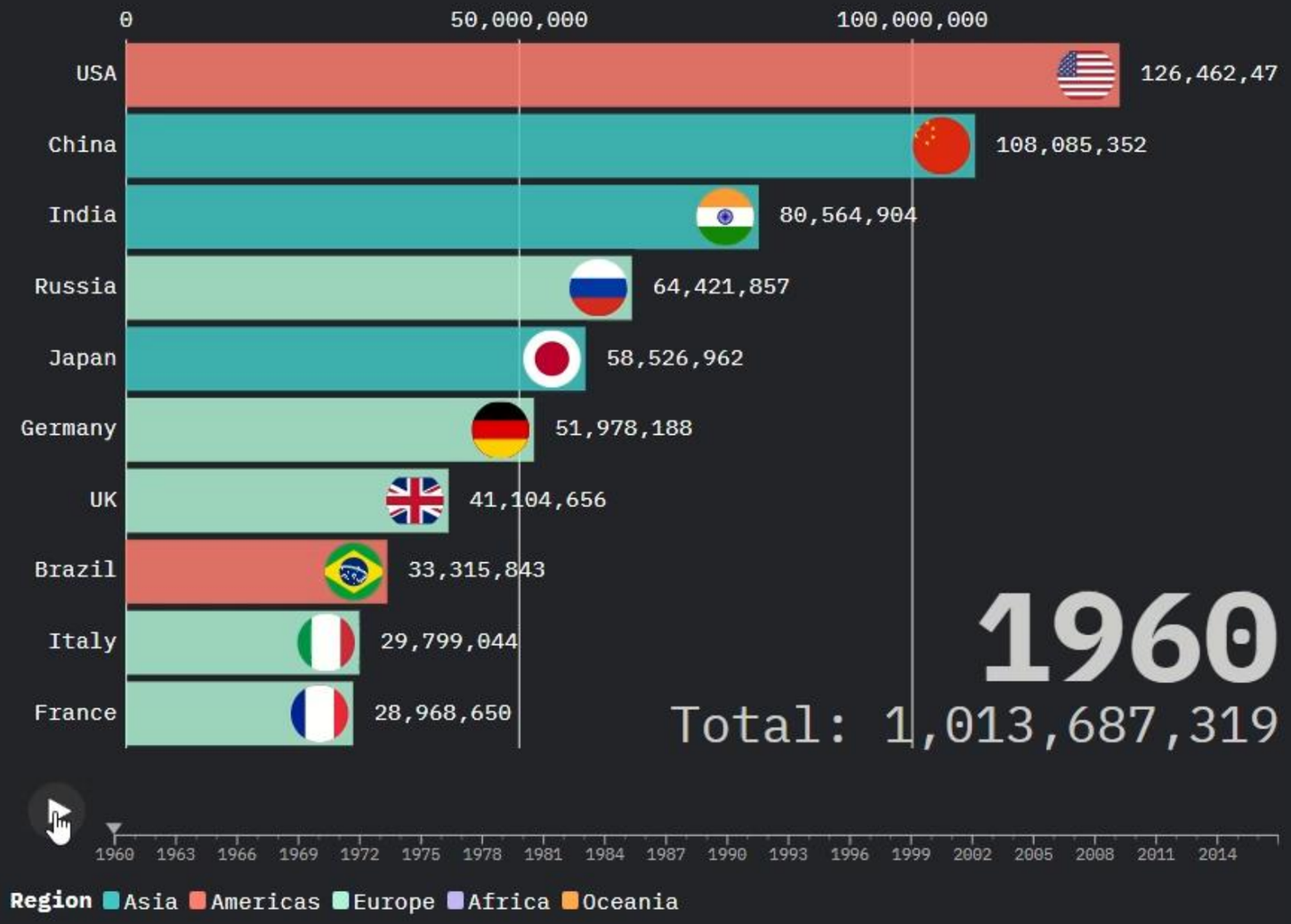


https://www.moonrakersdb.com/crew

Munzner's Rules of Thumb

- Eyes beat memory
  - Human short-term memory is extremely limited (≈ 4 "chunks" of info).
  - Visualizations can put information side-by-side, so users can compare at a glance instead of remembering values.
  - The eye can process spatial and visual relationships far more efficiently than the brain can store and recall numbers.

- Eyes beat memory **(Animation)**

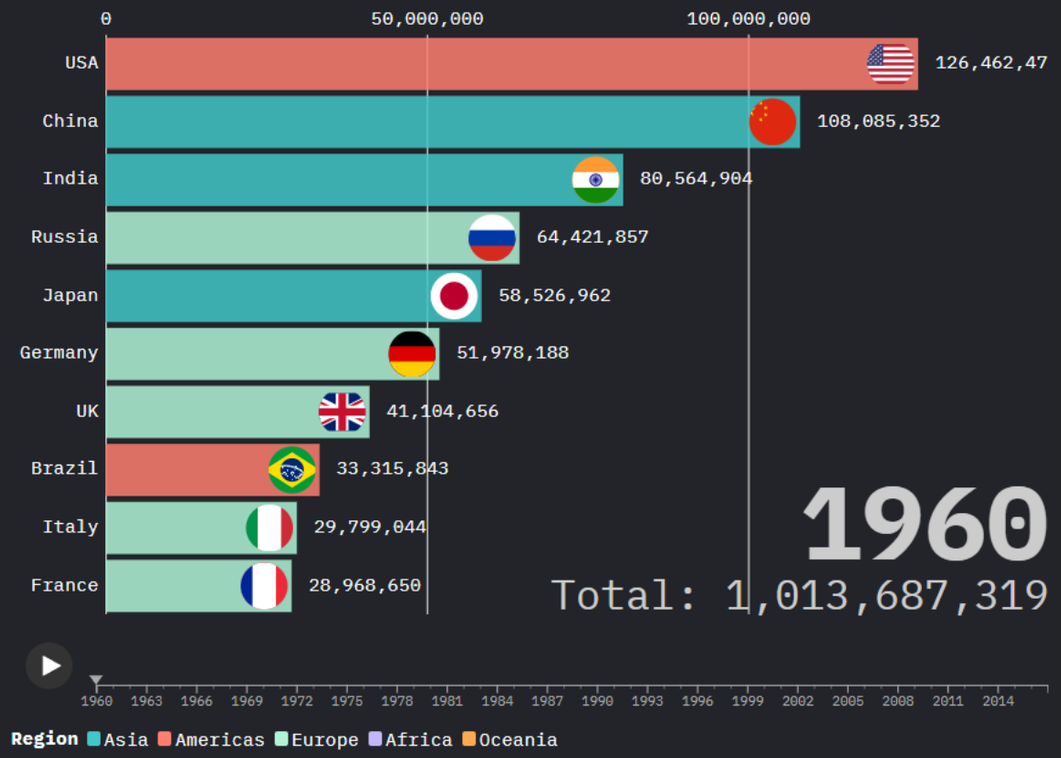


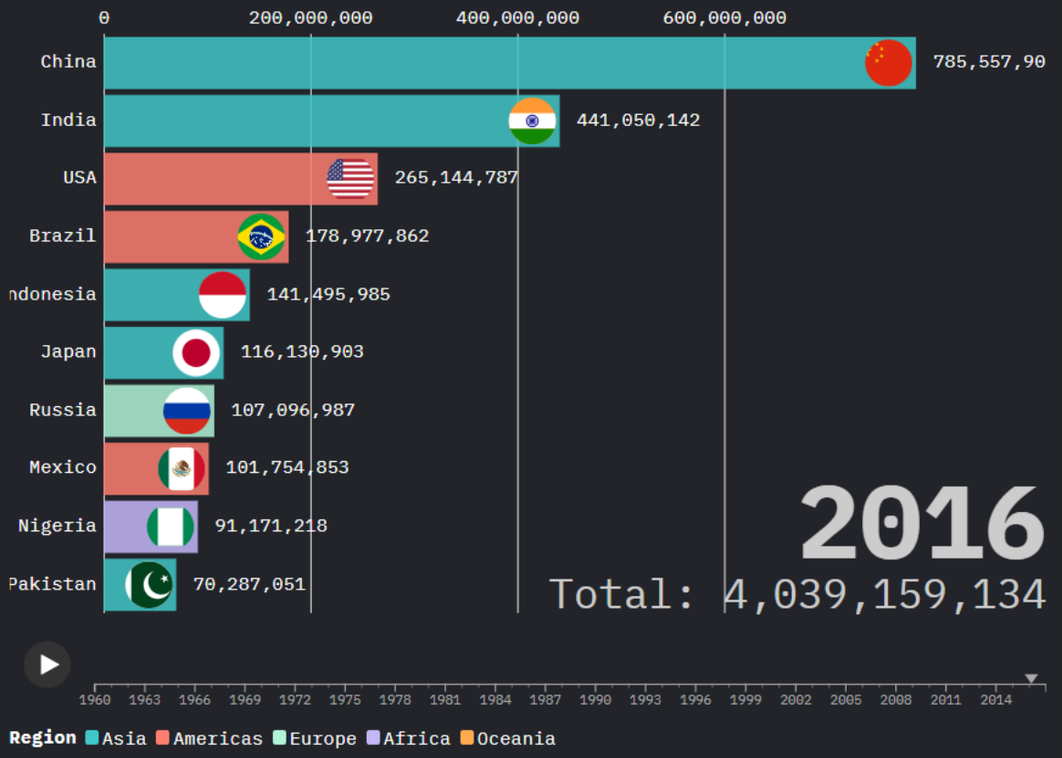
Changing of fortunes over time

- Which countries were in both the first and last years?
- Which countries were in first year that were not in the last year?
- Which region was predominant in 1960 or 2016?

Eyes beat memory **(Small Multiples)**

Changing of fortunes over time

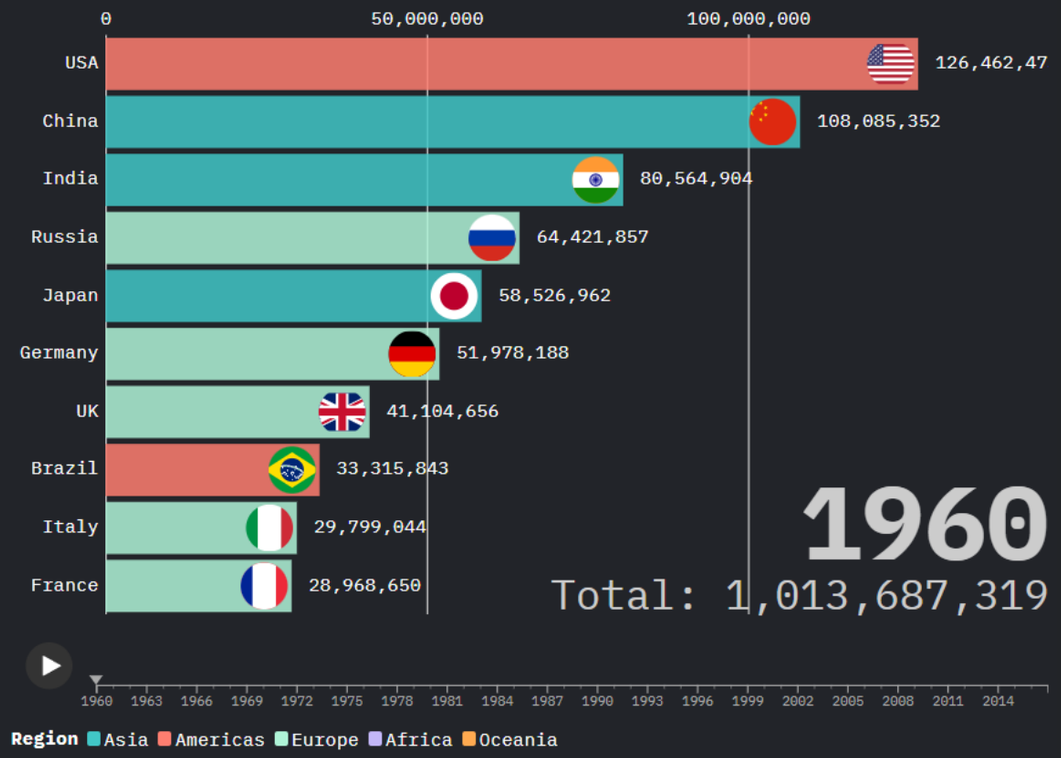


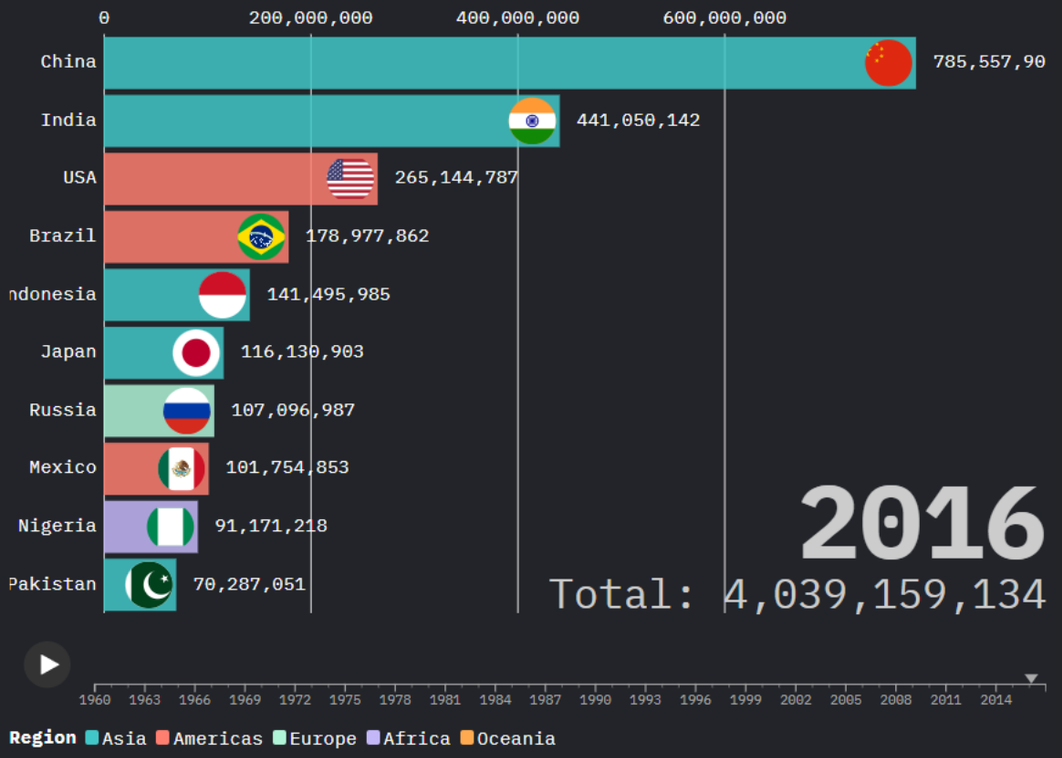


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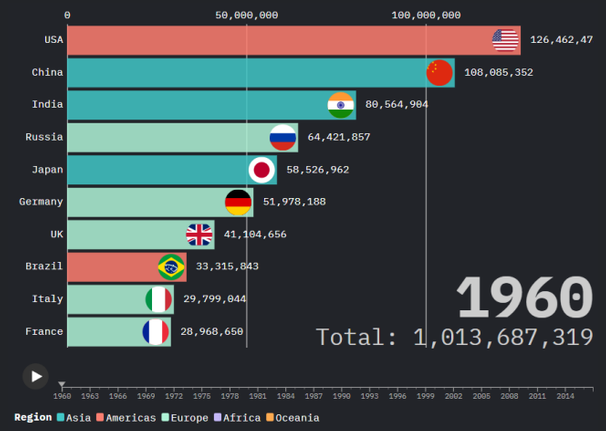


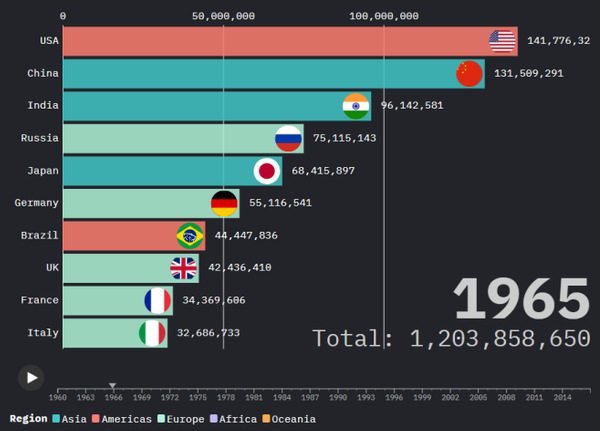


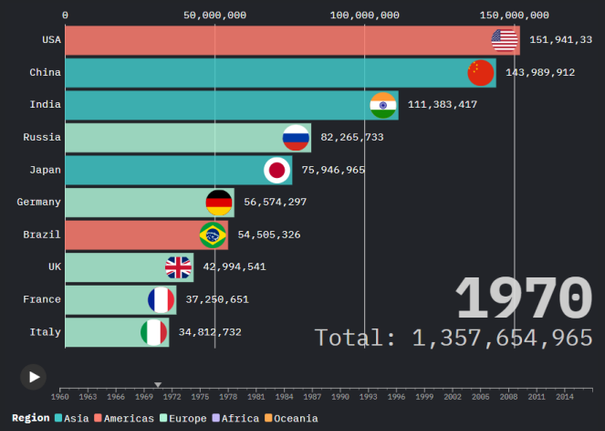
What is the problem of Small Multiples?

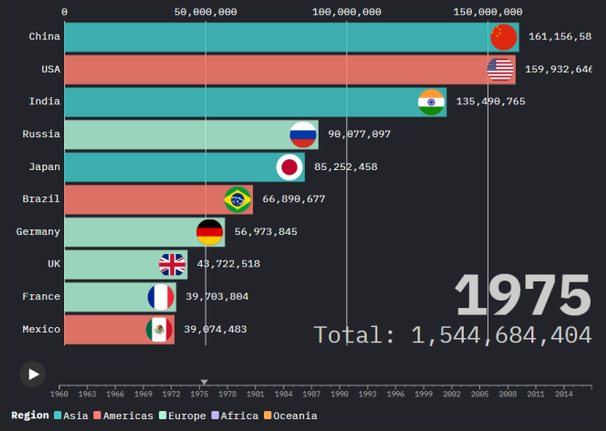
Eyes beat memory **(Small Multiples)**

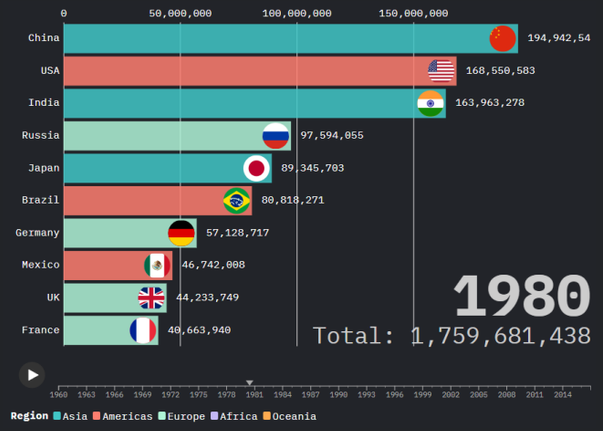
Changing of fortunes over time

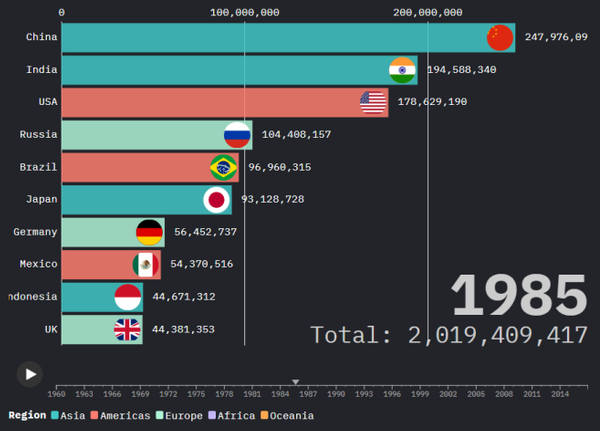


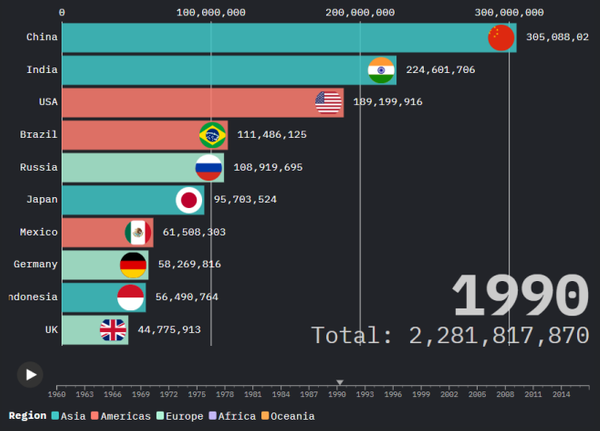


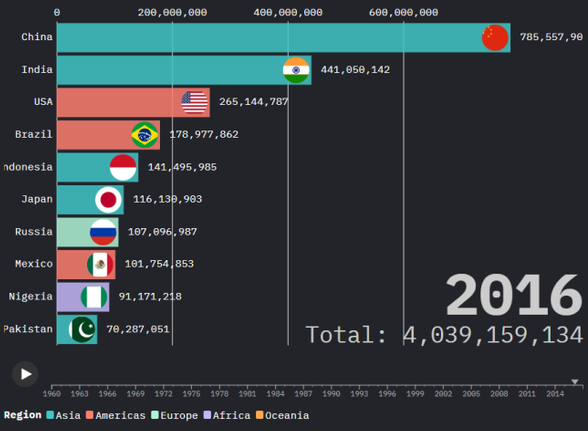






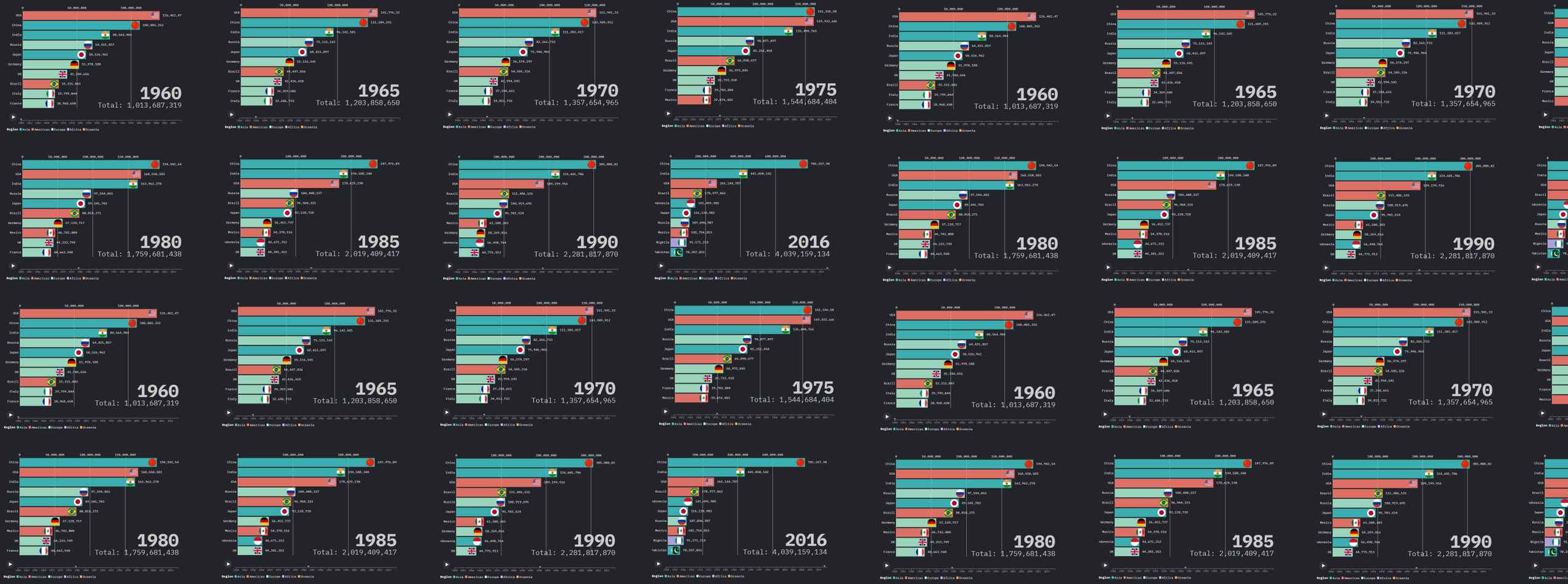




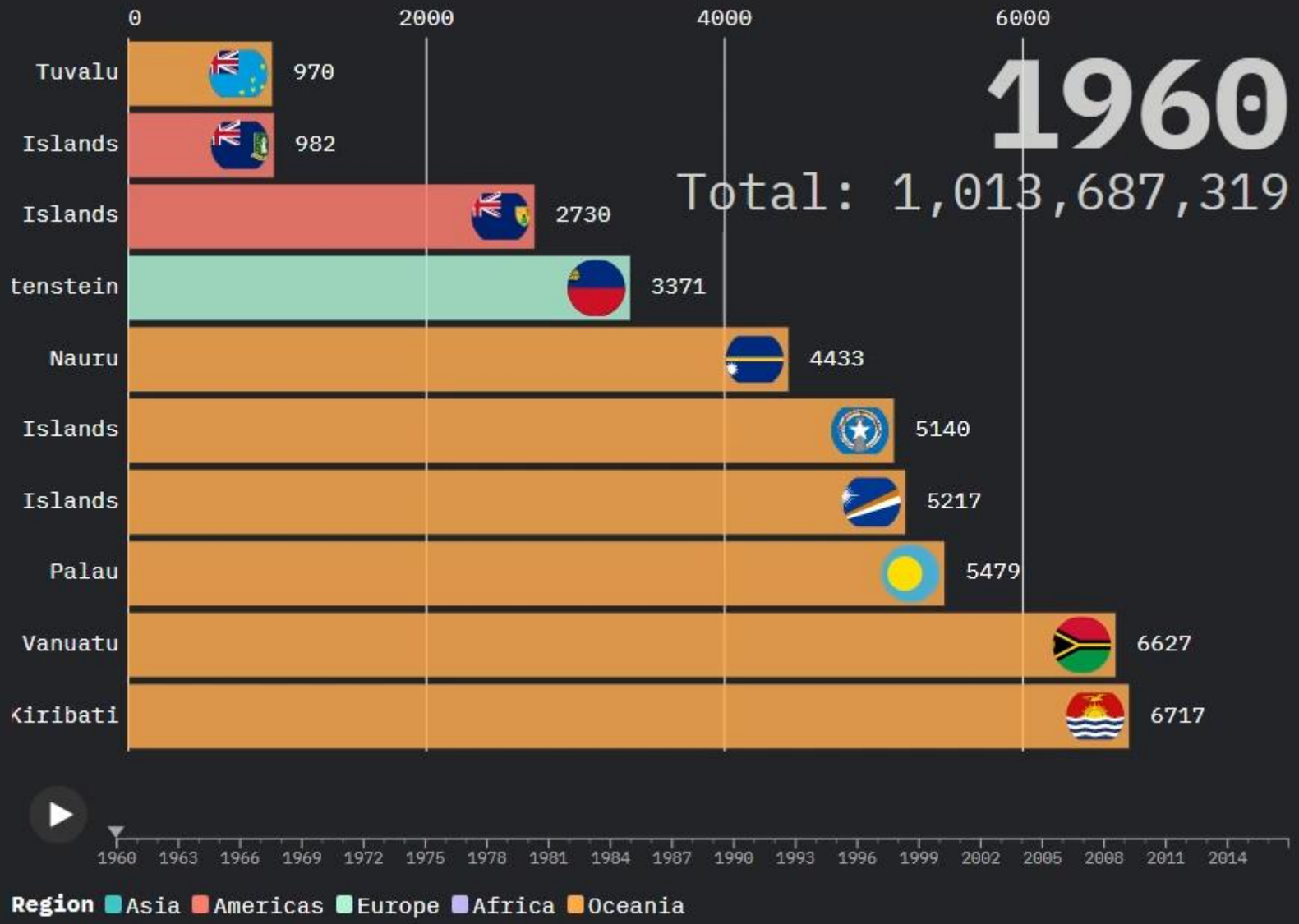


Eyes beat memory **(Small Multiples)**

Changing of fortunes over time

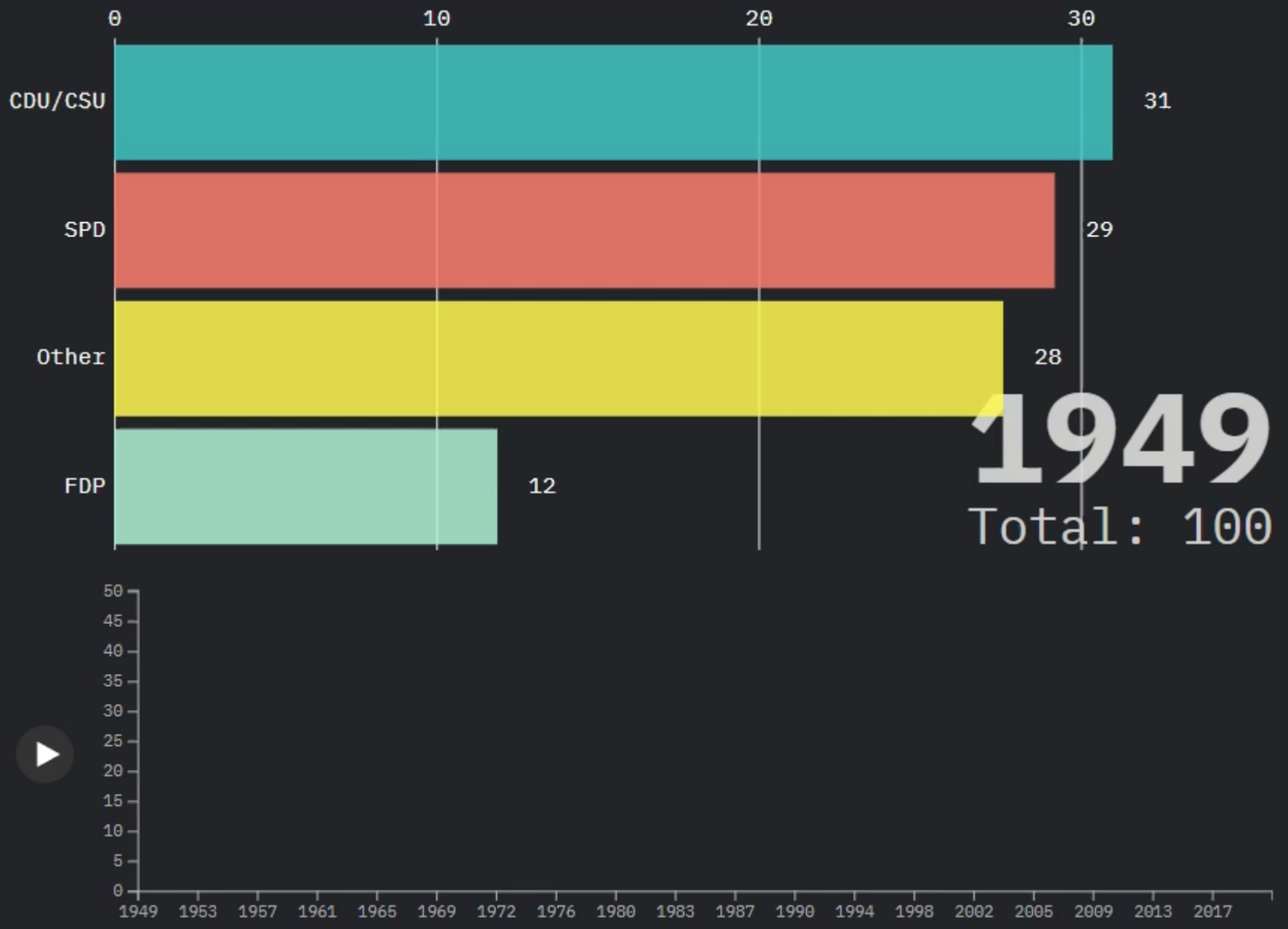


- Eyes beat memory



Animation with interaction needs to be used with precaution!

- Eyes beat memory



Multiple Coordinated Views (MCV)

- Ways to facet into multiple views: juxtaposition, integrated, and superimpose layers.

- Coordination options: linked selection, coordinated panning/zooming, shared legends/controls.

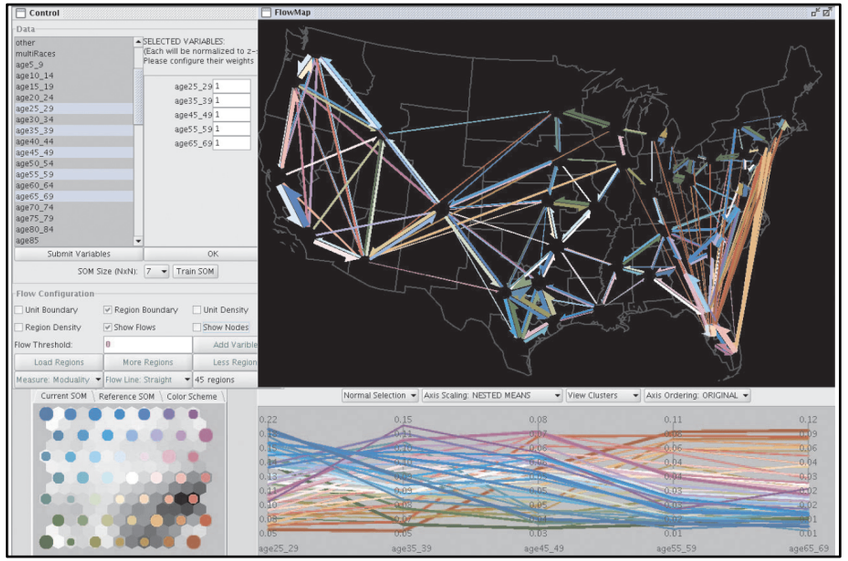
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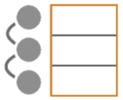
Juxtaposition: 

In the context of MCV visualization, juxtaposed views separate the attribute visualization into two or more views.

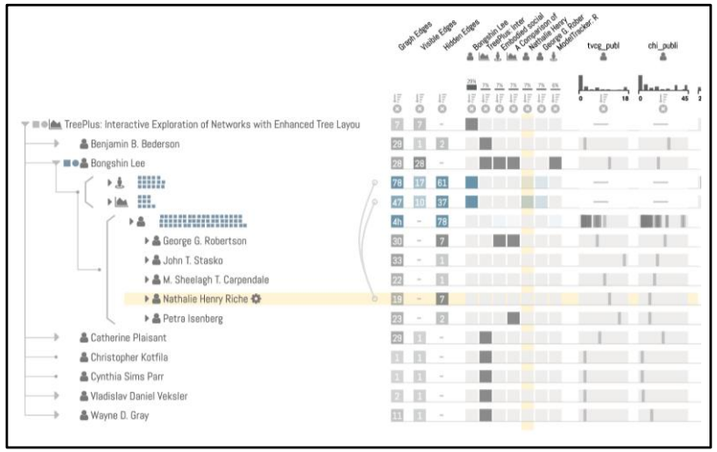


https://vdl.sci.utah.edu/mvnv/techniques/

Multiple Coordinated Views (MCV)

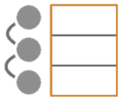
Integrated: 

Unlike juxtaposed views, in integrated views, the topology and the attribute visualizations are laid out with the other view in mind.

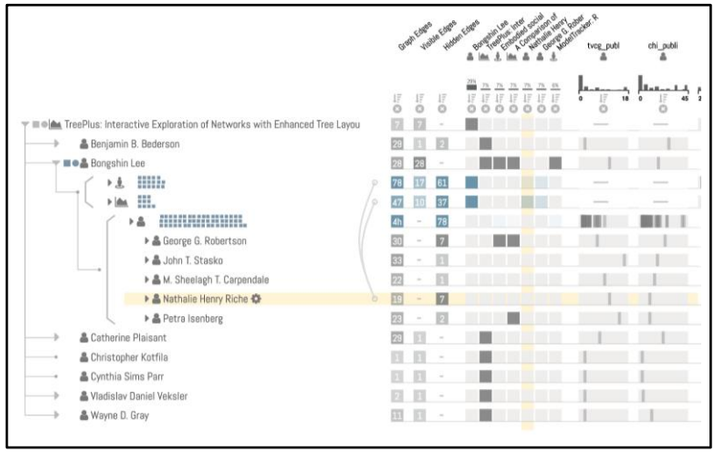


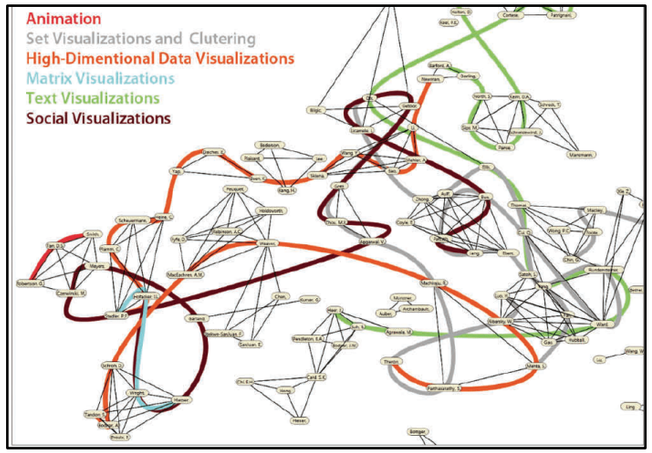
https://vdl.sci.utah.edu/mvnv/techniques/

Multiple Coordinated Views (MCV)

Integrated: 

Unlike juxtaposed views, in integrated views, the topology and the attribute visualizations are laid out with the other view in mind.





Superimpose: 

Superimposed views are those that display two encodings on top of each other.

https://vdl.sci.utah.edu/mvnv/techniques/

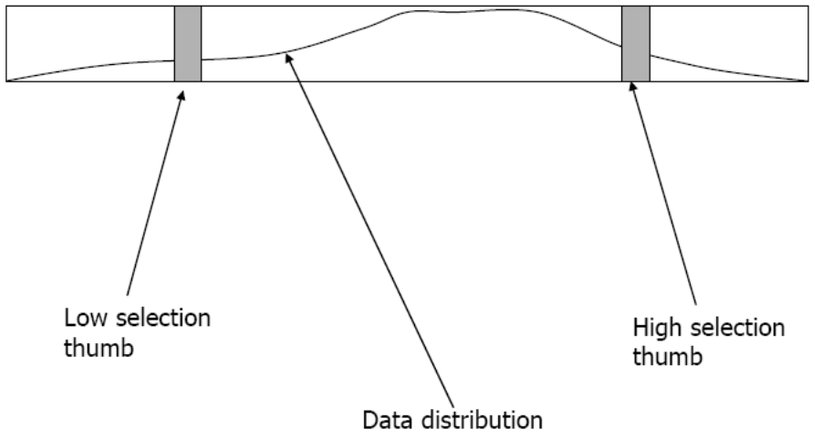
Dynamic Queries

- Dynamic queries are interactive filters that let users adjust query parameters (e.g., with sliders, checkboxes, buttons) and immediately see updated results in the visualization.

- Properties
  - Fast (visual) answer on the query, i.e., we get so-called “interactive answers” (≤ 0.1 seconds)
  - Simultaneous display of query and result
  - Exploration
  - Dynamic exploration, i.e., activities take place according to the “What if...?” principle

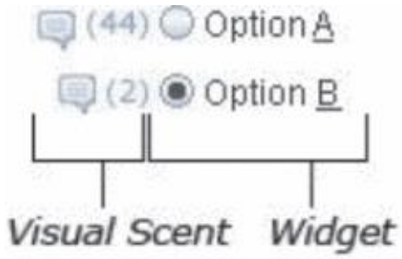
Dynamic Queries

- Data Visualization Sliders



- Scented Widgets

[Eick et al., UIST ´94]





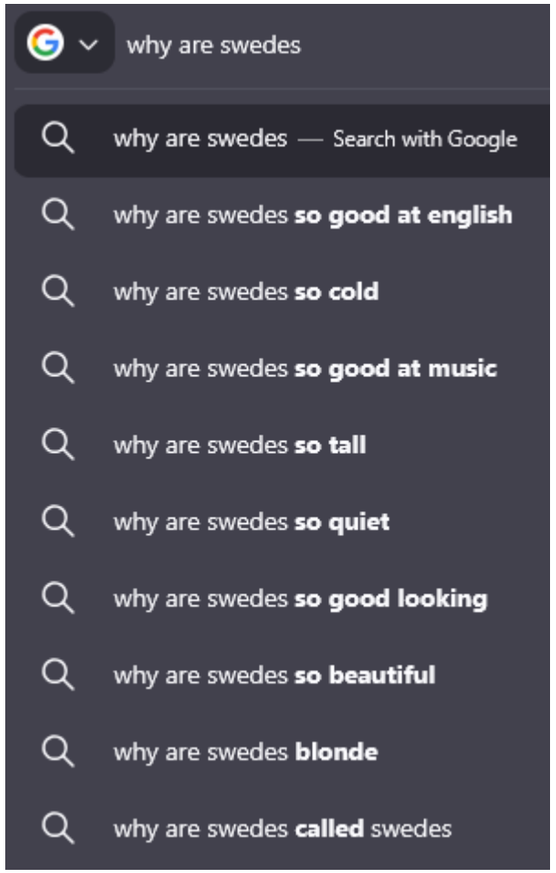
[Willett et al., InfoVis ´07]

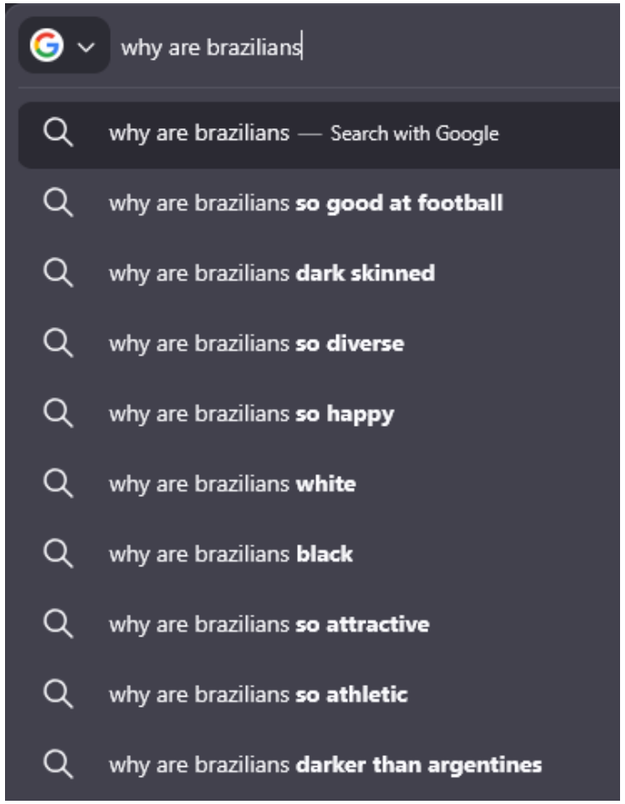
Dynamic Queries

- Range slider + Histogram



- Google search

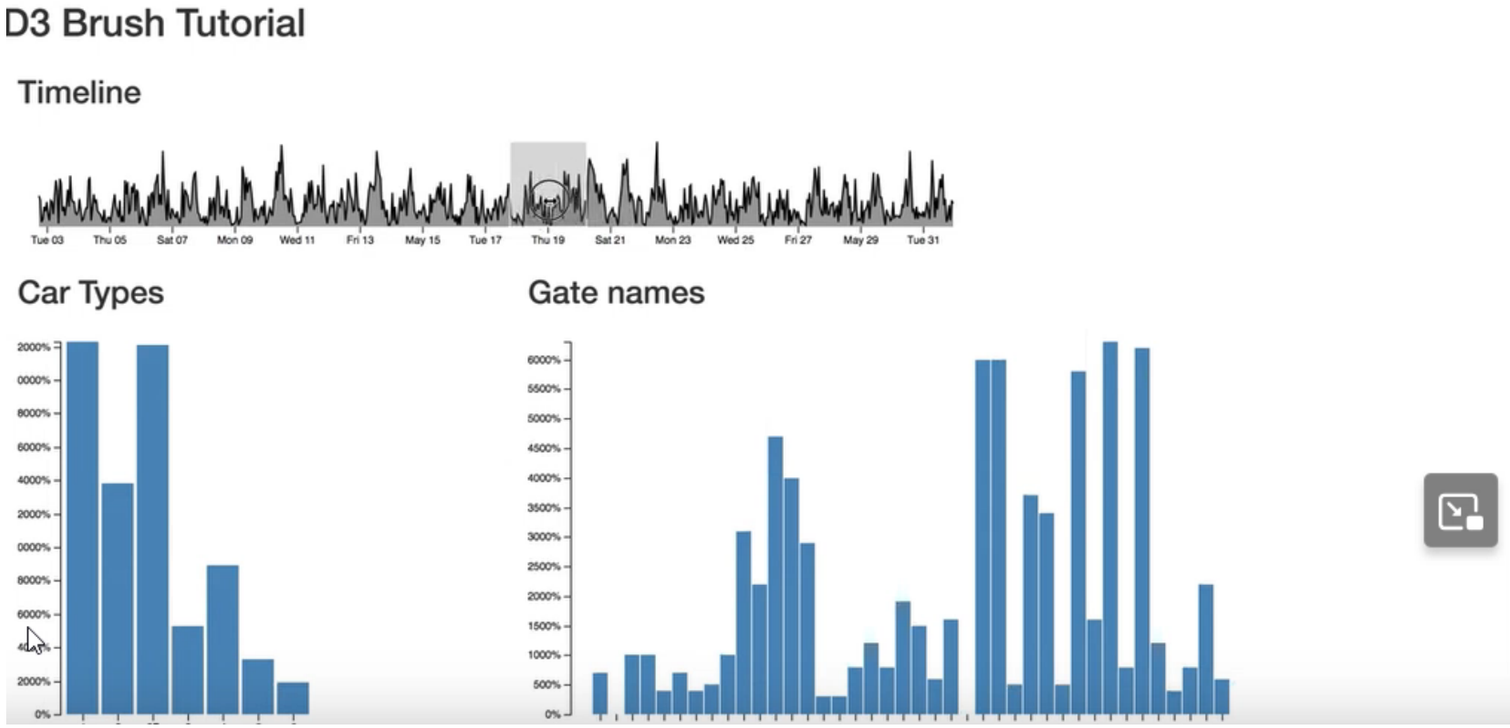




[Inspired by Bach's lecturer]

Brushing & Linking

- *Brushing & Linking* is used if several views on the same data object are needed
- Selecting or highlighting of a data object in a specific view (brushing) leads to a highlight/selection in another view (linking)
  - Consistent display
  - Facilitates the search



https://www.youtube.com/watch?v=Oz3U38oOcNg

Preserving the Mental Map

Interaction should support the user’s internal model of the system.

A strong mental map reduces confusion and helps users stay oriented.

[Inspired by Bach’s lecturer]

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  - Clear position indicators (zoom level, highlighted node, navigation aid).





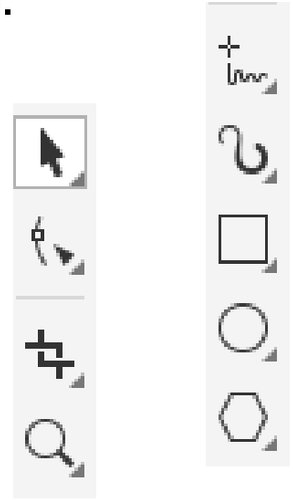
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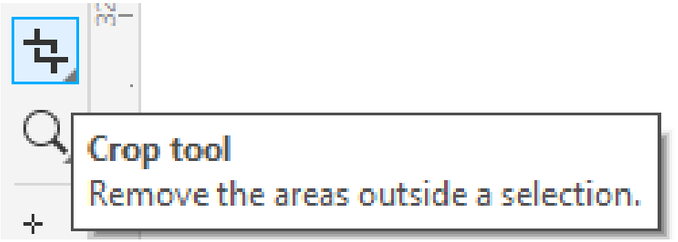
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- What can I do?
  - Make available interactions visible (filters, zoom, drag, brushing).
- How do I do that?
  - Intuitive, consistent interaction design (tooltips, buttons).



[Inspired by Bach's lecturer]

Guidelines for Interaction in Visualization

- Try static first
  - Start simple; add interaction when it improves insight.

[Inspired by Bach's lecturer]

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- Overview first, zoom & filter, detail on demand
  - Classic Shneiderman mantra.

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- Support user tasks
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- Overview first, zoom & filter, detail on demand
  - Classic Shneiderman mantra.
- Minimize interaction
  - Reduce clicks and steps → streamline workflow.

[Inspired by Bach's lecturer]

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  - Make controls and effects visible (e.g., tooltips, legends).

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- Provide direct manipulation
  - Users should interact directly with marks (e.g., brushing points).
- Consider people's devices
  - Design for desktop, tablet, mobile → input methods differ.

[Inspired by Bach's lecturer]

Further Reading

- **[Mun]** T. Munzner. Visualization Analysis and Design. 1st Edition, A K Peters/CRC Press, ISBN 978-1466508910, 2014.
  - Chapter 6: Rules of Thumb

